

Density estimation

Tasks (Lab 3):

1.
 - Generate a sample of size 200 from the mixture of two Gaussian distributions, i.e. from a density:

$$f(x) = 0.9 \cdot f_{N(5,1)}(x) + 0.1 \cdot f_{N(10,1)}(x).$$

- Use **kernel density estimator** to approximate the true, theoretical density f corresponding to the above distribution. Draw the density function corresponding to the theoretical distribution and the estimated function $\hat{f}_n$.
- Compute mean squared error (MSE)

$$\frac{1}{K} \sum_{i=1}^K [f(x_i) - \hat{f}_n(x_i)]^2,$$

where $x_1, \dots, x_K$ are iid. from $U([2, 12])$.

- Analyse how the error depends on sample size n . Generate a plot showing how the mean squared error changes with n .
 - Analyse the influence of different kernel functions as well as smoothing parameters. Generate curves for different kernels and different values of smoothing parameters. Try at least 3 kernels and 3 values of smoothing parameters. In addition, use one of the smoothing parameter estimation methods.
 - What is the effect of too small / too large bandwidth?
2. Generate sample $X_1, \dots, X_n$ of size $n = 200$ from Gaussian mixture model described above. Compare two methods (compute MSE for both of them):
 - (a) **Method 1:** Kernel density estimator using sample $X_1, \dots, X_n$
 - (b) **Method 2:** Using sample $X_1, \dots, X_n$, generate artificial sample $X'_1, \dots, X'_k$ (you can choose k much larger than n) corresponding to the kernel density as follows:
 - Generate i from uniform distribution on $\{1, \dots, n\}$
 - Generate ϵ from $N(0, 1)$
 - Set $X'_j = X_i + \epsilon \cdot h$

Compute kernel density estimator using modified sample $X'_1, \dots, X'_k$.

Interpret why the two methods should be close. Does increasing k change anything essential?

3. Select any dataset corresponding to binary classification problem with quantitative variables. Compare the accuracy of:

- Naive Bayes method (with kernel density estimator)
- Naive Bayes method (with Gaussian approximation of the density)
- Naive Bayes method (with discretization of quantitative features)
- LDA